

Sejoon Oh, Ph.D.

AI Foundation Research Scientist @ Netflix | EB-1B Green Card Holder
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RESEARCH SUMMARY

AI Researcher specializing in **Personalizing AI**, **Multimodal AI**, and **AI Safety and Robustness**. Proven track record of bridging the gap between research and production at Netflix, developing foundation models that drove **\$10M+ in Y-Y revenue lift** and optimized inference costs by **30%**. Expert in aligning LLMs via Post-training methods (DPO, GRPO, etc.), designing universal safety guardrails for Multimodal LLMs, and integrating semantic IDs into LLMs for hyper-personalized content discovery.

EXPERIENCE

Netflix

AI Foundation Research Scientist

Los Gatos, CA

June 2024 – Present

Multimodal & Personalized LLMs: Core researcher developing a multimodal LLM that ingests complex user engagement signals. Utilizing multi-task Supervised Fine-Tuning (SFT) and Reinforcement Learning (DPO, GRPO, etc.) to enhance personalization.

Semantic ID Integration: Leading the integration of Semantic IDs into LLMs, enabling the unified recommendation of diverse content types (videos, games, live events, podcasts), which are previously incompatible with standard ML models.

Foundation Model for User Intent (FM-Intent): Invented a hierarchical multi-task learning foundation model to **predict user session intent**. Successfully productized the model, achieving an impressive revenue lift Y/Y across millions of users.

Lightweight Foundation Models (FM-light): Developed a lightweight foundation model framework. Achieved **30% faster runtime** and **50% memory savings** while increasing recommendation freshness by 50% through dynamic response to users.

Discovery Optimization: Created a **specialized fine-tuning mechanism** for the Netflix foundation model focused on content discovery, resulting in a significant revenue lift Y/Y in global A/B testing across millions of users.

Netflix

Machine Learning Research Intern

Los Gatos, CA

May 2023 – Aug 2023

Project IntentRec: Developed "IntentRec" to predict user session intent using deep learning. Research laid the groundwork for the FM-Intent production model. Collaborated with mentors Dr. Moumita Bhattacharya & Dr. Yesu Feng.

The Home Depot

Data Science Research Intern

Atlanta, GA

May 2021 – Aug 2021

Real-time Recommendation: Researched and implemented "M2TRec," a metadata-aware multi-task transformer for session-based recommendations. Solved cold-start problems and improved real-time personalization accuracy.

Adobe Research

Data Science Research Intern

San Jose, CA

May 2020 – Aug 2020

Neural Tensor Completion: Developed influence-guided data augmentation techniques for neural tensor completion. Work resulted in a publication at CIKM 2021.

Georgia Institute of Technology

Graduate Research Assistant

Atlanta, GA

Aug 2019 – May 2024

Adversarial AI Safety: Developed "UniGuard," a universal safety guardrail to defend Multimodal LLMs against jailbreak attacks (AAAI 2025). Demonstrated vulnerabilities in text-aware recommender systems via adversarial text rewriting (CIKM 2024).

Recommender Systems: Authored multiple papers on stabilizing recommendations (TKDD 2024) and rank list sensitivity (CIKM 2022). Investigated cross-modal projection in Multimodal LLMs (ACL 2024).

Carnegie Mellon University

Graduate Research Assistant

Pittsburgh, PA

Aug 2018 – May 2019

Conducted research on machine learning approaches for computational biology problems.

Seoul National University

Undergraduate Research Intern, Data Mining Lab

Seoul, South Korea

July 2016 – May 2018

Researched Tensor Analysis, Recommender Systems, and High-Performance Computing.

EDUCATION

Georgia Institute of Technology

Ph.D. in Computer Science (Advisor: Prof. Srijan Kumar)

Focus: Multimodal AI, AI Safety, Recommender Systems, Adversarial ML.

Atlanta, GA

Aug 2019 – May 2024

Awards: Kwanjeong Educational Foundation Ph.D. Fellowship (\$30K/yr).

Seoul National University

B.S. in Computer Science and Engineering (Advisor: Prof. U Kang)

Awards: Best Thesis Award (1st in CSE), Humantech Paper Award (Gold Prize), National Science Scholarship.

Seoul, South Korea

Mar 2012 – Aug 2018

PATENTS

M. Bhattacharya, Y. Feng, S. Lamkhede, **S. Oh**. "Predicting User Session Intent with Hierarchical Multi-Task Learning." *Provisional Patent*.

W. Shalaby, **S. Oh**, A. H. Afsharinejad, X. Cui. "Machine Learning-based User Selection Prediction Based on Sequence of Prior User Selections." *Filed*.

S. Kim, **S. Oh**, R. Rossi. "Enhancing Neural-based Prediction of Multi-dimensional Data via Influence and Data Augmentation." *US Patent 20230244926A1*.

S. Oh, N. Park, U. Kang. "Apparatus for Supporting Multi-dimensional Data Analysis through Parallel Processing." *Korean Patent 10-2017-0158951*.

SELECTED PUBLICATIONS

Conference Papers

S. Oh, Y. Jin, M. Sharma, et al. "UniGuard: Towards Universal Safety Guardrails for Jailbreak Attacks on Multimodal Large Language Models." *Deployable AI Workshop at AAAI*, 2025.

S. Oh, G. Verma, S. Kumar. "Adversarial Text Rewriting for Text-aware Recommender Systems." *ACM CIKM*, 2024.

G. Verma... **S. Oh**, et al. "Cross-Modal Projection in Multimodal LLMs Doesn't Really Project Visual Attributes to Textual Space." *ACL*, 2024.

S. Oh, J. McAuley, B. Ustun, S. Kumar. "Rank List Sensitivity of Recommender Systems to Interaction Perturbations." *ACM CIKM*, 2022.

S. Oh, S. Kim, R. Rossi, S. Kumar. "Influence-guided Data Augmentation for Neural Tensor Completion." *ACM CIKM*, 2021.

S. Oh, N. Park, L. Sael, U. Kang. "Scalable Tucker Factorization for Sparse Tensors." *IEEE ICDE*, 2018.

Journal Papers

S. Oh, J. McAuley, B. Ustun, S. Kumar. "FINEST: Stabilizing Recommendations by Rank-Preserving Fine-Tuning." *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 2024.

S. Oh, J. Lee, L. Sael. "GIFT: Guided and Interpretable Factorization for Tensors..." *Bioinformatics*, 2018.

Technical Blogs

S. Oh, M. Bhattacharya, Y. Feng, S. Lamkhede. "FM-Intent: Predicting User Session Intent with Hierarchical Multi-Task Learning." *Netflix Tech Blog*, 2025.

PROFESSIONAL SERVICES

Conference Reviewer: AAAI 2024, KDD 2023, CIKM 2021-2022.

Journal Reviewer: ACM Transactions on Knowledge Discovery from Data (TKDD), IEEE TPDS.

TECHNICAL SKILLS

Core AI: GenAI, Foundation Models, Multimodal LLMs, Post-training, Recommender Systems, Adversarial Attack and Defense.

Languages & Frameworks: Python, PyTorch, TensorFlow, HuggingFace, C, Scala, TorchTune, Qwen, Llama

Techniques: LLM, VLM, Multi-task Learning, Semantic IDs, Model Distillation, High-Performance Computing, Tensor Analysis.

HONORS & AWARDS

EB-1B Green Card Holder (Outstanding Professors and Researchers)

Jan 2025

Kwanjeong Ph.D. Fellowship (\$150K total funding, awarded to top Korean students)

2019 – 2024

Best Thesis Award (Seoul National University, 1st among all CSE undergrads)

2018

Humantech Paper Award (Gold Prize, Samsung, 1st in Computer Science)

2018